

AD Security Lab

Troubleshooting & Pro Tips

Your Quick Reference for Common Issues

Common Issues & Solutions

Problem: VM Won't Start / Crashes

- **Symptoms:**
 - VMware shows 'Not enough physical memory'
 - VM crashes during boot
 - Host computer becomes very slow
- **Solutions:**
 - Check available RAM: Close other programs, browsers, etc.
 - Reduce VM RAM allocation: DC and Win11 can run with 2GB each (slower but works)
 - Don't run all VMs at once: Start DC first, then Win11 when needed
 - Increase host paging file: System Properties → Advanced → Performance Settings
 - Restart VMware Workstation completely

Windows: Check RAM usage in Task Manager (Ctrl+Shift+Esc)

Problem: Can't Join Domain

- **Symptoms:**
 - 'Domain company.local could not be contacted'
 - 'The specified domain either does not exist or could not be contacted'
 - 'DNS name does not exist'
- **Solutions (Try in order):**
 1. Verify DNS is set correctly on WIN11-CLIENT:
 - *Preferred DNS: 192.168.100.10 (the DC)*
 2. Test connectivity from WIN11-CLIENT PowerShell:

```
ping 192.168.100.10  
ping company.local  
nslookup company.local
```

3. Make sure DC1 is fully booted and running
4. Check both VMs are on VMnet2 network
5. On DC1, verify DNS service is running:

```
Get-Service DNS
```

6. Disable Windows Firewall temporarily (for testing):

```
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled  
False
```

Problem: Splunk Not Receiving Data

- **Symptoms:**

- Search shows 'No results found'
- Can't see events from WIN11-CLIENT
- Splunk shows 0 events

- **Solutions:**

1. Wait 5-10 minutes (initial indexing takes time)

2. Check Splunk services on DC1:

```
Get-Service splunkd
Get-Service SplunkForwarder # if installed
```

3. Verify receiving port 9997 is open on DC1:

*Settings → Forwarding and Receiving → Configure receiving
Should show port 9997 enabled*

4. Check forwarder on WIN11-CLIENT:

```
cd "C:\Program Files\SplunkUniversalForwarder\bin"
.\splunk.exe list forward-server
```

5. Open firewall on DC1 for Splunk:

```
New-NetFirewallRule -DisplayName "Splunk Receiving" `
-Direction Inbound -Protocol TCP -LocalPort 9997 -Action
Allow
```

6. Restart Splunk services on both machines

Problem: High CPU Usage / Computer Slow

- **Symptoms:**

- Computer fans running loud
- Host computer very slow
- VMware using 80-100% CPU

- **Solutions:**

- This is NORMAL when Splunk first starts indexing (30-60 minutes)
- Let it finish! It will calm down after initial indexing
- Reduce data inputs if too slow (disable Application logs)
- Power off VMs when not using them
- Use 'Suspend' instead of 'Shutdown' for faster restarts

Problem: Login Failed / Can't Remember Password

The standard password used throughout: P@ssw0rd123!

- **Accounts:**

- DC1 Local Admin: Administrator / P@ssw0rd123!
- DC1 Domain Admin: COMPANY\Administrator / P@ssw0rd123!
- Domain User: COMPANY\unnathi.vithlani / P@ssw0rd123!
- WIN11 Local: LocalUser / P@ssw0rd123!
- Splunk: admin / P@ssw0rd123!

 IMPORTANT:

These are LAB passwords only! NEVER use simple passwords like this in production environments!

Pro Tips & Best Practices

Snapshot Your VMs!

CRITICAL: Take snapshots after each major step!

-  After DC1 is promoted to domain controller
-  After WIN11-CLIENT joins domain
-  After Sysmon is installed
-  After Splunk is fully configured
-  Before running any attacks

How to snapshot: VM → Snapshot → Take Snapshot → Give it a descriptive name

Speed Up Your Workflow

- ⚡ **Use Suspend:** Instead of shutting down VMs, use 'Suspend'. Resumes in seconds!
- ⚡ **Clone VMs:** After basic setup, clone WIN11-CLIENT to create multiple clients quickly
- ⚡ **Shared Folders:** Enable VMware shared folders to easily transfer files
- ⚡ **Copy/Paste:** Install VMware Tools for copy/paste between host and VMs
- ⚡ **Multiple Desktops:** Use Windows virtual desktops (Win+Tab) to organize your screens

Organize Your Screenshots

Create a folder structure for documentation:

```
Screenshots/  
├── 01_VMware_Setup/  
├── 02_DC1_Installation/  
├── 03_WIN11_Setup/  
├── 04_Active_Directory/  
├── 05_Splunk_Dashboard/  
├── 06_Detections/  
└── 07_Attacks/
```

Make it Portfolio-Ready

-  Add comments to your Splunk queries explaining the logic
-  Create a network diagram showing your architecture
-  Document WHY each detection is important (business context)
-  Include false positive analysis in your GitHub
-  Add 'Lessons Learned' section to your documentation
-  Screenshot your detections with real alerts triggered



Network Architecture Diagram

Here's what your lab architecture looks like:

Component	IP Address	Role / Services
VMnet2	192.168.100.0/24	Isolated Host-Only Network
DC1	192.168.100.10	Domain Controller, DNS, Splunk Enterprise
WIN11-CLIENT	192.168.100.20	Domain-joined Workstation, Splunk Forwarder, Sysmon
Kali Linux	192.168.100.30	Attack Platform (optional)
Your Host	192.168.100.1	Can access all VMs via VMnet2



Traffic Flow:

- WIN11-CLIENT → DC1 (port 389/636): LDAP authentication
- WIN11-CLIENT → DC1 (port 53): DNS queries
- WIN11-CLIENT → DC1 (port 9997): Splunk log forwarding
- Kali → WIN11/DC1 (various): Attack traffic
- You → DC1 (port 8000): Access Splunk web interface

Additional Detection Ideas

After you complete the basic lab, try implementing these detections:

1. Unusual Logon Times

What it detects: Detect logins outside business hours (nights/weekends)

SPL Query:

```
index=main EventCode=4624 | eval hour=strftime(_time, "%H") | where hour<6 OR hour>20
```

2. Multiple Failed Logins

What it detects: Brute force detection (5+ failures in 5 minutes)

SPL Query:

```
index=main EventCode=4625 | stats count by Account_Name | where count>5
```

3. New User Creation

What it detects: Alert when new AD users are created

SPL Query:

```
index=main EventCode=4720 | table _time, Account_Name, Creator
```

4. Privilege Escalation

What it detects: Detect when users are added to Admin groups

SPL Query:

```
index=main EventCode=4728 OR EventCode=4732 | rex field=Group_Name "Admin"
```

5. Suspicious PowerShell

What it detects: Detect encoded PowerShell commands

SPL Query:

```
index=main EventCode=4104 | search "encodedcommand" OR "bypass"
```

6. Port Scanning

What it detects: Detect network scanning activity

SPL Query:

```
index=main EventCode=5156 | stats count by dest_ip | where count>50
```

Learning Resources

Free Training & Certifications

- **Splunk Fundamentals:** Free course on Splunk.com
https://www.splunk.com/en_us/training.html
- **MITRE ATT&CK:** Learn attack techniques
<https://attack.mitre.org>
- **Active Directory Basics:** Microsoft Learn
<https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/>
- **Sysmon Config:** SwiftOnSecurity GitHub
<https://github.com/SwiftOnSecurity/sysmon-config>
- **Detection Engineering:** Florian Roth's GitHub
<https://github.com/Neo23x0/>
- **Threat Hunting:** ThreatHunter-Playbook
<https://threathunterplaybook.com>

YouTube Channels

-  John Hammond - Security content & CTFs
-  IppSec - HackTheBox walkthroughs
-  NetworkChuck - Networking & security
-  The Cyber Mentor - Pentesting & hacking
-  13Cubed - DFIR & forensics

Next Steps After Completion

Expand Your Lab

- **Add More Clients:** Create 2-3 more Windows clients to simulate real network
- **Linux Server:** Add Ubuntu server for web applications
- **Attacker Tools:** Install Metasploit, Cobalt Strike, etc. on Kali
- **Email Server:** Set up Exchange or mail server for phishing simulations
- **Network Segmentation:** Create DMZ, internal network zones
- **Vulnerability Scanner:** Add Nessus or OpenVAS
- **Honeypot:** Deploy honeypot to attract attacks

Advanced Attack Scenarios

-  Pass-the-Hash attacks
-  Golden Ticket creation (Mimikatz)
-  NTLM Relay attacks
-  Bloodhound AD enumeration
-  Responder poisoning
-  Privilege escalation chains
-  Lateral movement simulations

Build Your Brand

- **LinkedIn Post:** Share your lab with screenshots and learnings
- **Blog Series:** Write detailed setup guides on Medium or dev.to
- **YouTube Video:** Record a demo of your lab and detections
- **Conference Talk:** Present at local BSides or security meetups
- **Open Source:** Contribute detection rules to SIGMA or other projects
- **Portfolio Website:** Create a security.yourname.com site
- **Mentorship:** Help others build their labs (pay it forward!)



You're Building Something Great!

This lab is more than just a technical exercise – it's your entry ticket into cybersecurity careers.

Hiring managers LOVE candidates who:

- **Show initiative by building their own labs**
- **Can demonstrate hands-on skills**
- **Understand both attack and defense**
- **Document their work professionally**

Your GitHub + LinkedIn + This Lab = Interview Calls

Keep building, keep learning, and don't hesitate to reach out to the community for help!

Remember: Every security professional started exactly where you are now.

The difference? They started.

You've got this! 💪